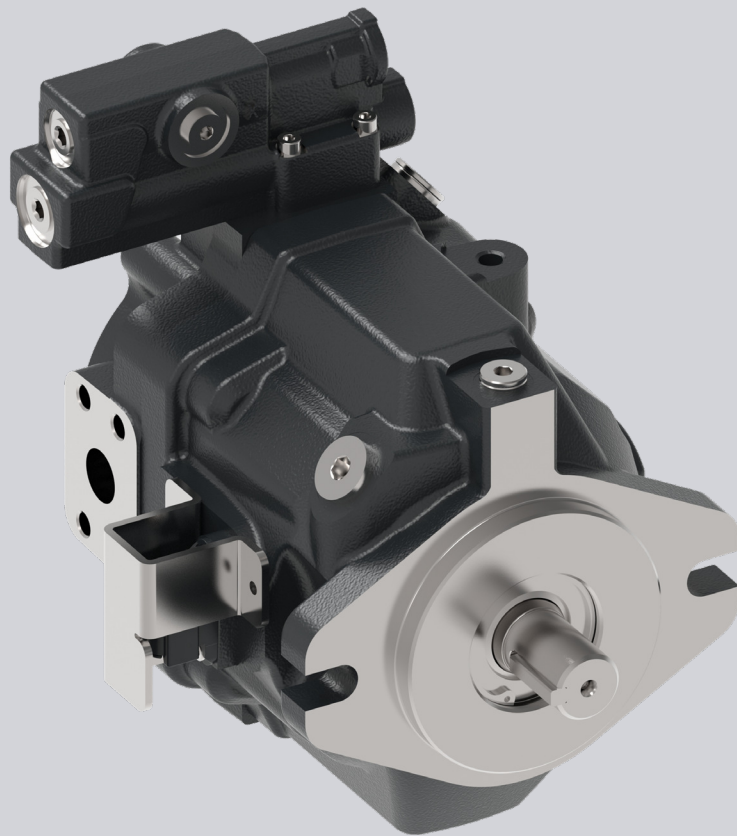




Vickers® by Danfoss

PVMX Open Circuit Axial Piston Pump



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1. Introduction

1.1 Using this manual

This manual includes information for the normal operation, maintenance, and service of the PVMX open circuit variable displacement pumps. The manual includes a description of the units and their individual components, troubleshooting information, adjustment instructions and minor repair procedures.

Many service and adjustment activities can be performed without removing the unit from the machine. However, adequate access to the unit must be available, and the unit must be thoroughly cleaned before beginning maintenance, adjustment, or repair activities. Since dirt and contamination are the greatest enemies of any type of hydraulic equipment, follow cleanliness requirements strictly. This is especially important when changing the system filter and when removing hoses or plumbing.

1.2 Safety Precautions

Always consider safety precautions before beginning a service procedure. Protect yourself and others from injury. Take the following general precautions whenever servicing a hydraulic system.

Unintended machine movement

WARNING

Unintended movement of the machine or mechanism may cause injury to the technician or bystanders. To protect against unintended movement, secure the machine or disable / disconnect the mechanism while servicing.

Flammable cleaning solvents

WARNING

Some cleaning solvents are flammable. To avoid possible fire, do not use cleaning solvents in an area where a source of ignition may be present.

Fluid under pressure

WARNING

Escaping hydraulic fluid under pressure can have sufficient force to penetrate your skin causing serious injury and/or infection. This fluid may also be hot enough to cause burns. Use caution when dealing with hydraulic fluid under pressure. Relieve pressure in the system before removing hoses, fittings, gauges, or components. Never use your hand or any other body part to check for leaks in a pressurized line. Seek medical attention immediately if you are cut by hydraulic fluid.

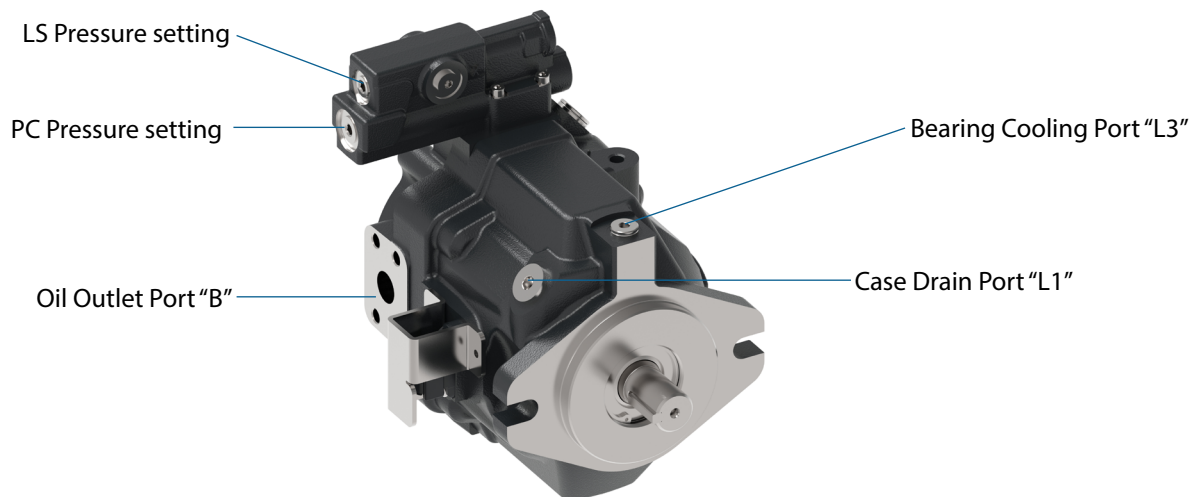
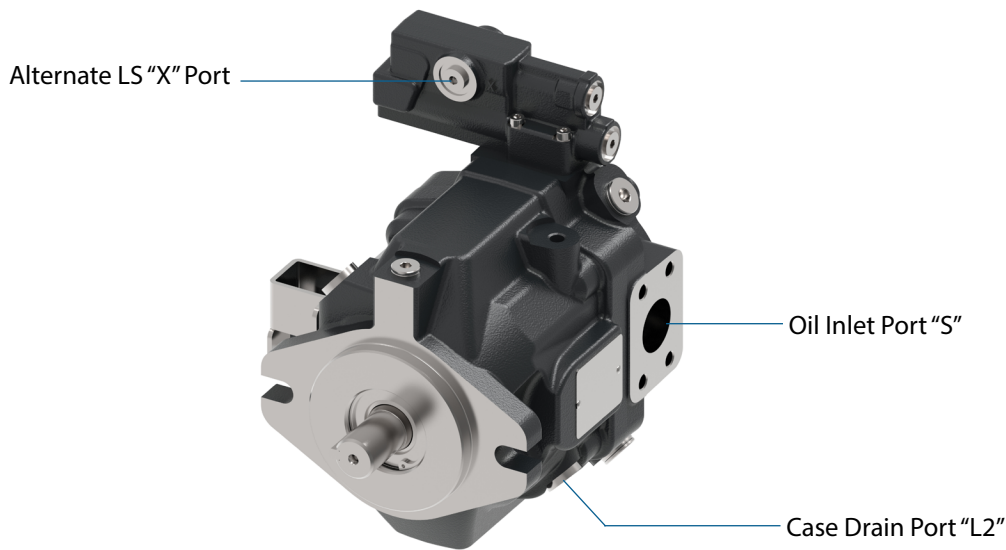
Personal safety

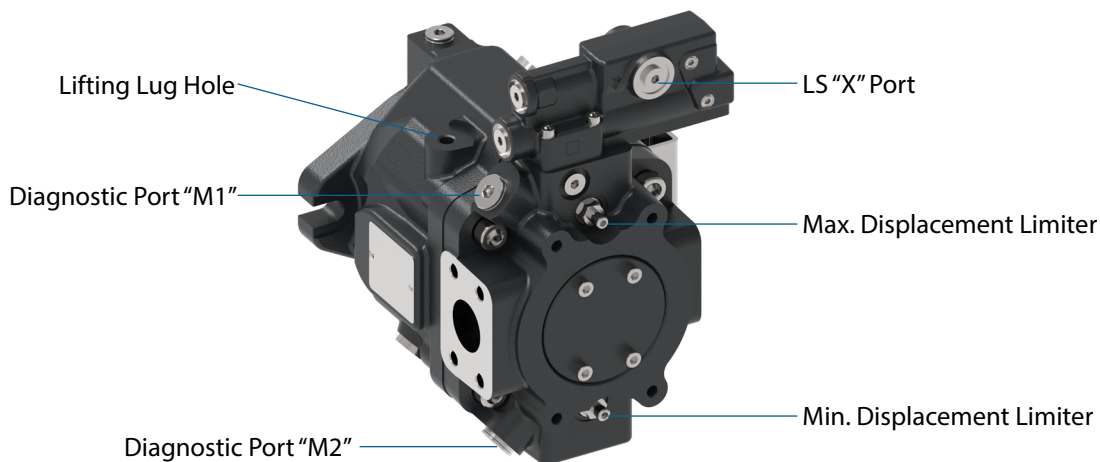
WARNING

Protect yourself from injury. Use proper safety equipment, including safety glasses, at all times.

1.3 General description

The PVMX is an open-circuit, variable displacement axial piston pump that converts rotational mechanical power from a prime mover into hydraulic power. The input drive shaft is supported by taper roller bearings and is connected to the cylinder block via splines. This cylinder block contains nine reciprocating pistons, each fitted with a slipper that slides on an angled swashplate. The swashplate angle, which determines the pump's output flow, is controlled by two opposing forces. A bias piston and spring mechanism pushes the swashplate towards an inclined position, while the servo piston acts against it. When hydraulic pressure in the servo cylinder increases, it overcomes the spring force and reduces the swashplate angle, thereby de-stroking the pump. A variety of control strategies are available which are described later. A stationary valve plate directs the fluid from the suction side to the pressure side as the cylinder block rotates. A small, controlled amount of internal leakage between the cylinder block-valve plate and slipper-swashplate interfaces provides essential lubrication and cooling for the internal components. This fluid is then collected and returned to the reservoir through the case drain ports.





Service Port Options			
Inlet/Outlet Port Option	Port Code	Inlet Port "S"	Outlet Port "B"
Inch Tube	1	SAE J514 O-ring-20, 1 5/8-12 UNF-2B	SAE J514 O-ring-12, 1 1/16-12 UNF-2B
Inch Flange	2	SAE J518 Code 61, standard pressure. 1-1/4 inch diameter, 3/8-16 UNC-2B	SAE J518 Code 62, high pressure. 3/4 inch diameter, 3/8-16 UNC-2B
Metric Tube	3	ISO 6149-1, M42 thread	ISO 6149-1, M27 thread
Metric Flange	4	ISO 6162-1 32mm diameter, M10x25 bolt holes	ISO 6162-2 20mm diameter, M10x20 bolt holes

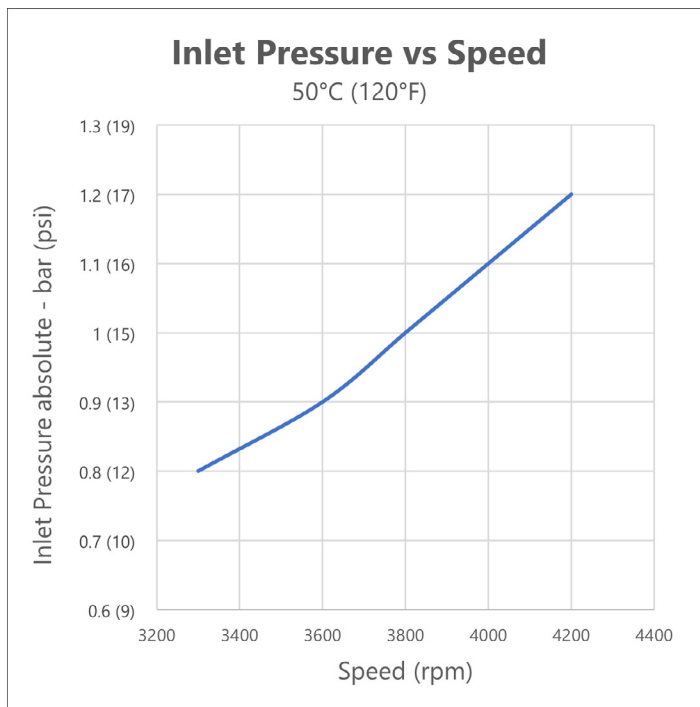
Drain Ports, Load Sensing Ports, Bearing Flushing Port Options				
Inlet/Outlet Port Option	Port Code	Drain Port "L1&L2"	Load Sensing Port "X"	Bearing Flush Port "L3"
Inch Flange or Tube	1, 2	SAE J1926/1:3/4-16	ISO 11926-1 - 7/16-20 UNF-2B	SAE J1926/1:1/2-20
Inch Flange	3, 4	ISO 6149-1 O-ring M18x1.5 thread	ISO 6149-1 O-ring M12x1.5 thread	ISO 6149-1 O-ring M12x1.5 thread

Mounting Flange Options						
2-bolt or 4-bolt flange	Code	Flange Description	A	E	F	J
2-bolt	B	SAE J744-101-2	Ø14.3	73	146	101.57±0.025
2-bolt	M	ISO 3019/2-100A2HW	Ø13.5	70	140	99.975±0.025

2. Technical Specifications

Features	Unit/Type	PVMX028
Maximum displacement	cm ³ [in ³]	28 [1.71]
Minimum input speed	rpm	200
Nominal input speed	rpm	3800*
Maximum input speed	rpm	4200*
Flow at nominal speed (theoretical)	l/min [US gal/min]	106.4 [292.28]
Nominal pressure	bar [psi]	350 [5076]
Maximum pressure	bar [psi]	420 [6092]
Input power at nom. input speed and nom. pressure	kW [HP]	62.1 [83.2]
Input torque at max displacement and nom. pressure	N•m/bar [lbf•in/1000 psi]	156 [1380]
Mass moment of inertia of internal rotating components	kg•m ² [slug•ft ²]	1.483x10 ⁻³ [1.09x10 ⁻³]
Weight	kg [lb]	23 [50.7]

* - Refer to Inlet Pressure vs. Speed chart



3. Hydraulic Parameters

3.1 Inlet Pressure

Minimum (continuous)	0.8 bar absolute	6.7 in. Hg vac.
Minimum (cold start)	0.5 bar absolute	15.1 in. Hg vac.

3.2 Case Pressure

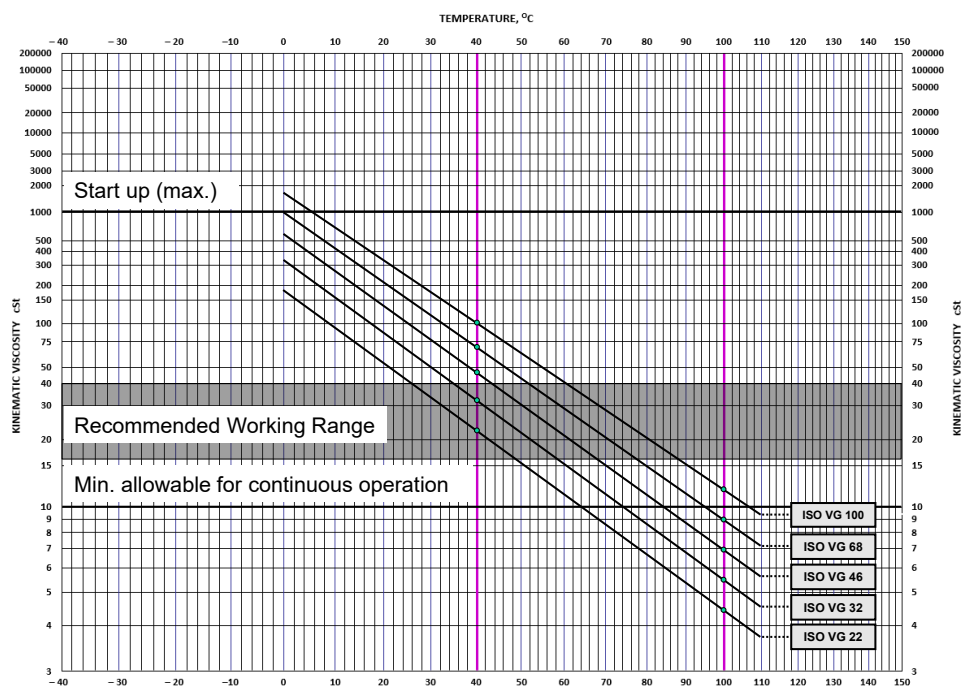
Maximum (continuous)	0.5 bar above inlet	7 psi above inlet
Maximum (cold start)	2 bar above inlet	29 psi above inlet

3.3 Temperature range

Minimum (intermittent, cold start)	- 40° C	- 40° F
Continuous	82° C	180° F
Maximum Intermittent	104° C	220° F

3.4 Fluid Viscosity

Condition		mm/s ² (cSt)	SUS
Minimum	Continuous	16	83
	Intermittent	10	90
Maximum	Continuous	40	187
	Intermittent	1000	4550



3.5 Hydraulic fluid

PVMX is designed to operate with HLP mineral oil according to DIN 51524. Pumps can also be operated with Fire resistance fluids with derated pressure and speed conditions.

3.6 Filtration

To prevent damage to the pump, including premature wear, fluid entering the pump inlet must be free of contaminants. PVMX pumps require system filtration capable of maintaining fluid cleanliness at ISO 4406-1999 class 22/18/13 or better. Danfoss does not recommend suction line filtration. Suction line filtration can cause high inlet vacuum, which limits pump operating speed. Instead, a 125 µm (150 mesh) screen in the reservoir covering the pump inlet is recommended. This protects the pump from coarse particle ingestion. Return line filtration is the preferred method for open circuit systems. Consider these factors when selecting a system filter:

- Cleanliness specifications
- Contaminant ingress rates
- Flow capacity
- Desired maintenance interval

Typically, a filter with a beta ratio of $\beta_{10} = 10$ is adequate. However, because each system is unique, only a thorough testing and evaluation program can fully validate the filtration system.

3.7 Reservoir

The reservoir provides clean fluid, dissipates heat, and removes entrained air from the hydraulic fluid. It allows for fluid volume changes associated with fluid expansion and cylinder differential volumes. Minimum reservoir capacity depends on the volume needed to perform these functions. Typically, a capacity of one to three times the pump flow (per minute) is satisfactory. Locate the reservoir outlet (suction line) near the bottom, allowing clearance for settling foreign particles. Place the reservoir inlet (return lines) below the lowest expected fluid level, as far away from the outlet as possible

3.8 Fluid velocity

Choose piping sizes and configurations sufficient to maintain optimum fluid velocity, and minimize pressure drops. This reduces noise, pressure drops, and overheating. It maximizes system life and performance.

Recommended fluid velocities:

System lines	6 to 9 m/s	20 to 30 ft/s
Suction lines	1 to 2 m/s	4 to 6 ft/s
Case drain	3 to 5 m/s	10 to 15 ft/s

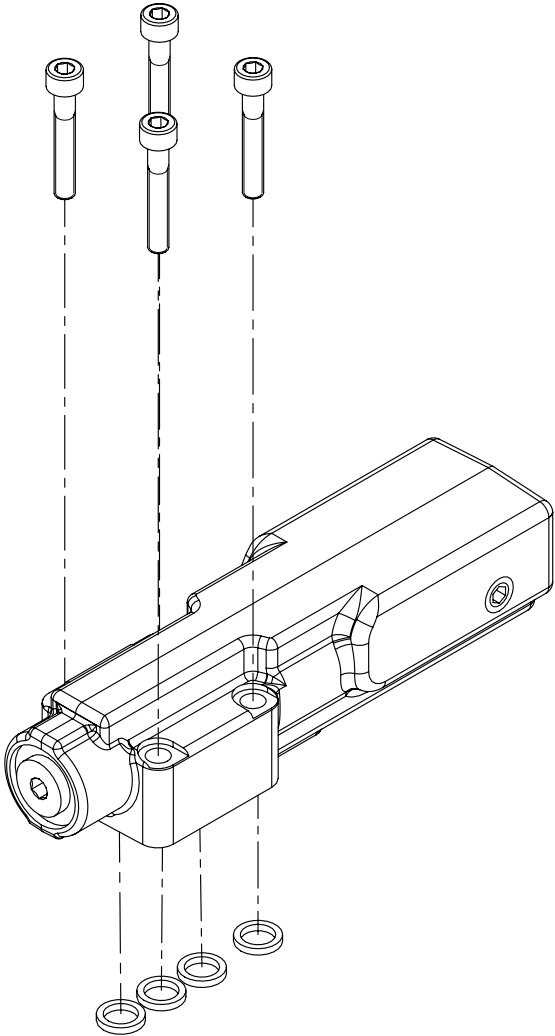
4. Controller Options

To meet the needs of various applications, PVMX features a comprehensive suite of control options. This includes traditional controllers, such as the Pressure Compensator and Load Sensing controller, as well as advanced electro-hydraulic solutions.

4.1 Pressure Compensator

The pump provides a continuously modulated flow to meet changing load demands at a preadjusted compensator pressure. At pressures below the compensator setting, the pump operates at maximum displacement.

Ordering Code	Kit description	PC Setting range	Pump size
			28 cc
11416223	Controller kit	60bar - 100 bar	●
11416091	Controller kit	100 bar - 280 bar	●
11416427	Controller kit	280 bar - 310 bar	●
11416435	Controller kit	310 bar - 350 bar	●
11421922	Controller seal kit	-	●



4.1.1 Pressure compensator setting

Pressure compensator setting is mentioned in the modelcode at position 15. The available options and the allowable setting range is mentioned in the table below.

Pressure compensator setting	
00	NONE
A0	070-100bar range 070±4 bar [1015±58 lbf/in ²]- Default setting
B0	100-280bar range 250±4 bar [1015±58 lbf/in ²]- Default setting
C0	280-310bar range 300±4 bar [1015±58 lbf/in ²]- Default setting
D0	310-350bar range 350±4 bar [1015±58 lbf/in ²]- Default setting

1. Install a pressure gauge measure pump outlet pressure. Install a pressure gauge in case drain port to measure case pressure.

WARNING

Unintended movement of the machine or mechanism may cause injury to the technician or bystanders. To protect against unintended movement, secure the machine or disable / disconnect the mechanism while servicing.

CAUTION

Contamination can damage internal components and void the manufacturer's warranty. Take precautions to ensure system cleanliness when removing and reinstalling system lines.

2. Start the prime mover and allow fluid to reach normal operating temperature. Operate a hydraulic function to its full extension, loading the pump at maximum pressure and zero flow.
3. Loosen the PC set screw and turn the PC adjustment screw until the desired setting is indicated on the pump outlet pressure gauge ¹⁾. Clockwise rotation will increase pressure; counterclockwise rotation will decrease; approximate gain = 35 bar [507 psi] per turn.

If the pressure does not increase, an external system relief valve may require adjustment. External system relief valve must be set above the PC setting for proper operation.

4. While holding the position of the PC adjustment screw, torque the PC set screw to 9 N·m [7 lbf·ft].
5. Stop the prime mover, remove the pressure gauges, and return the system to its normal operating configuration.

¹⁾ PC setting is referenced to case pressure. Subtract case pressure from system pressure to compute the actual setting.

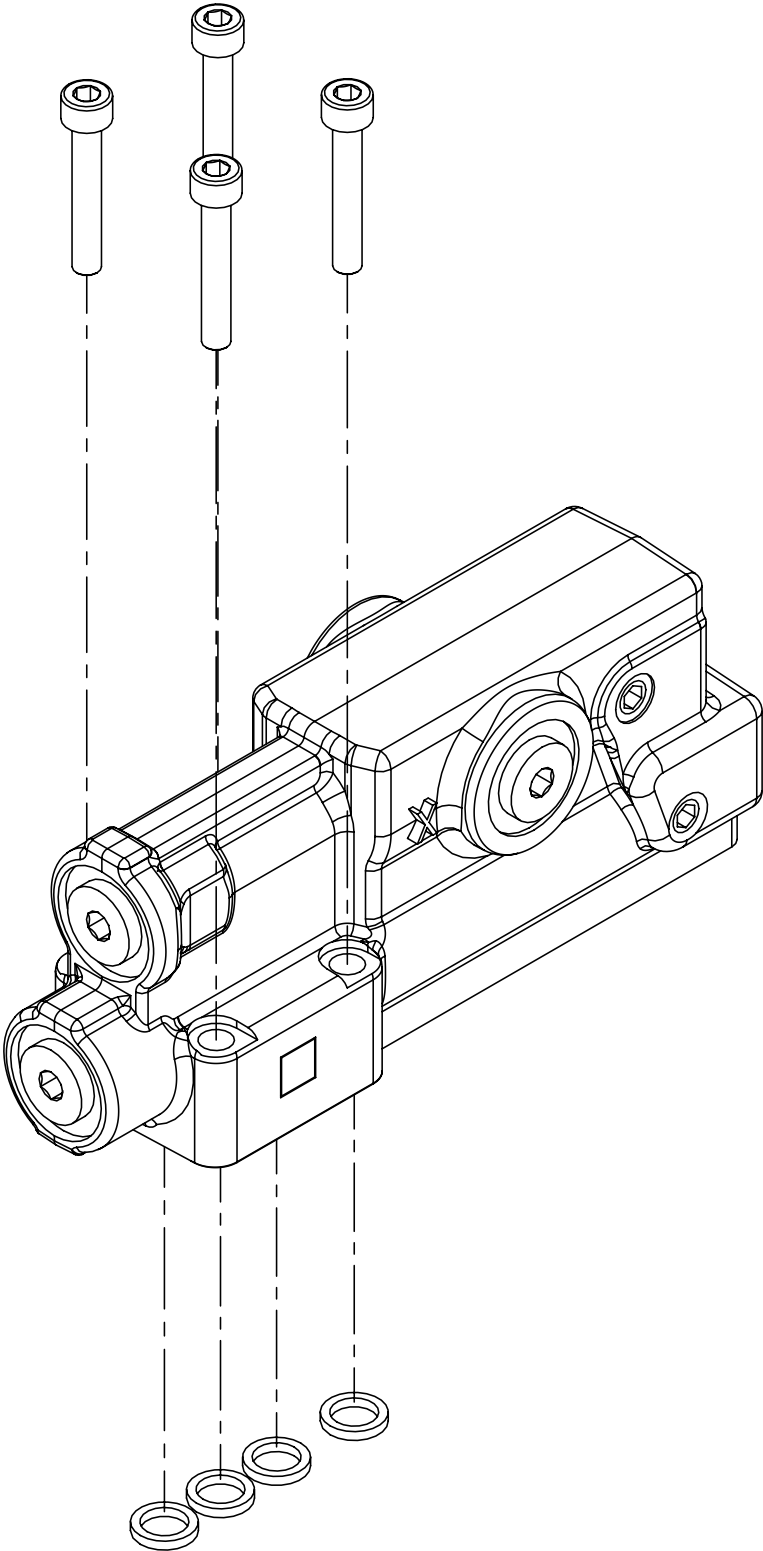
4.2 Pressure and Flow Compensator (Load sensing controller)

A load sensing (LS) control in an axial piston pump is an intelligent control system that automatically adjusts pump output based on actual system demand by maintaining a constant pressure differential between pump outlet pressure and the highest load pressure in the system. This is achieved through a control mechanism consisting of a load sensing line that reads system pressure, a control piston adjusting the swashplate angle, and a spring-loaded compensator spool. When an actuator demands flow, its directional valve opens, communicating the load pressure back to the pump control through the LS line, which then adjusts pump displacement to maintain the preset pressure differential. This results in energy-efficient operation since the pump only produces the required flow and pressure, reduces heat generation, allows multiple actuator operation without interference, and ensures low power consumption during standby mode when the pump operates at minimal flow.

LS control with bleed orifice

The load sense signal line requires a bleed orifice to prevent high-pressure lockup of the pump control. Most load-sensing control valves include this orifice. An optional internal bleed orifice is available, for use with control valves that do not internally bleed the LS signal to tank.

Ordering Code	Kit description	PC Setting range	LS Setting range	LS Port type	Bleed down orifice	Pump size
						28 cc
11421693	Controller kit	60bar - 100 bar	10 bar - 40 bar	SAE	Yes	●
11416546	Controller kit	100 bar - 280 bar	10 bar - 40 bar	SAE	Yes	●
11421681	Controller kit	280 bar - 310 bar	10 bar - 40 bar	SAE	Yes	●
11421705	Controller kit	310 bar - 350 bar	10 bar - 40 bar	SAE	Yes	●
11421707	Controller kit	60bar - 100 bar	10 bar - 40 bar	ISO	Yes	●
11421696	Controller kit	100 bar - 280 bar	10 bar - 40 bar	ISO	Yes	●
11421697	Controller kit	280 bar - 310 bar	10 bar - 40 bar	ISO	Yes	●
11421715	Controller kit	310 bar - 350 bar	10 bar - 40 bar	ISO	Yes	●
11421700	Controller kit	60bar - 100 bar	10 bar - 40 bar	SAE	No	●
11421720	Controller kit	100 bar - 280 bar	10 bar - 40 bar	SAE	No	●
11421745	Controller kit	280 bar - 310 bar	10 bar - 40 bar	SAE	No	●
11421753	Controller kit	310 bar - 350 bar	10 bar - 40 bar	SAE	No	●
11421748	Controller kit	60bar - 100 bar	10 bar - 40 bar	ISO	No	●
11421762	Controller kit	100 bar - 280 bar	10 bar - 40 bar	ISO	No	●
11421764	Controller kit	280 bar - 310 bar	10 bar - 40 bar	ISO	No	●
11421736	Controller kit	310 bar - 350 bar	10 bar - 40 bar	ISO	No	●
11421871	Controller seal kit	-	-	SAE	-	●
11421924	Controller seal kit	-	-	ISO	-	●



4.2.1 Pressure compensator and Flow compensator setting

Pressure and Flow compensator setting is mentioned in the modelcode at position 15 and 16 respectively. The available options and the allowable setting range is mentioned in the table below.

Pressure compensator setting	
00	NONE
A0	070-100bar range 070±4 bar [1015±58 lbf/in ²]- Default setting
B0	100-280bar range 250±4 bar [1015±58 lbf/in ²]- Default setting
C0	280-310bar range 300±4 bar [1015±58 lbf/in ²]- Default setting
D0	310-350bar range 350±4 bar [1015±58 lbf/in ²]- Default setting

Flow compensator setting	
0	NONE
A	10bar-40bar (DEFAULT 20±1bar [290±14.5 lbf/in ²])

1. Install a pressure gauge to measure pump outlet pressure. Install a pressure gauge in drain port to measure case pressure. Tee-in a gauge to the LS signal line.

WARNING

Escaping hydraulic fluid under pressure can have sufficient force to penetrate your skin causing serious injury and/or infection. Relieve pressure in the system before removing hoses, fittings, gauges, or components.

Unintended movement of the machine or mechanism may cause injury to the technician or bystanders. To protect against unintended movement, secure the machine or disable / disconnect the mechanism while servicing.

CAUTION

Contamination can damage internal components and void the manufacturer's warranty. Take precautions to ensure system cleanliness when removing and reinstalling system lines.

2. Start the prime mover and allow fluid to reach normal operating temperature. Slowly operate a hydraulic function which will demand approximately half flow from the pump but keep system pressure below the PC set point.
3. Loosen the LS set screw. While watching the pressure gauges, turn the LS adjustment screw until the desired pressure differential between the pump out pressure gauge and LS port is achieved. Clockwise rotation will increase the setting; counterclockwise rotation will decrease; approximate gain = 12 bar [170 psi] per turn.
4. While holding the position of the LS adjustment screw, torque the LS set screw to 9 N·m [7 lbf·ft].
5. Operate a hydraulic function to its full extension loading the pump at maximum pressure and zero flow.
6. Loosen the PC set screw and turn the PC adjustment screw until the desired setting is indicated on the pump outlet pressure gauge. Clockwise rotation will increase pressure; counterclockwise rotation will decrease; approximate gain = 35 bar [507 psi] per turn ¹⁾.

If the pressure does not increase, an external system relief valve may require adjustment. External system relief valve must be set above the PC setting for proper operation.

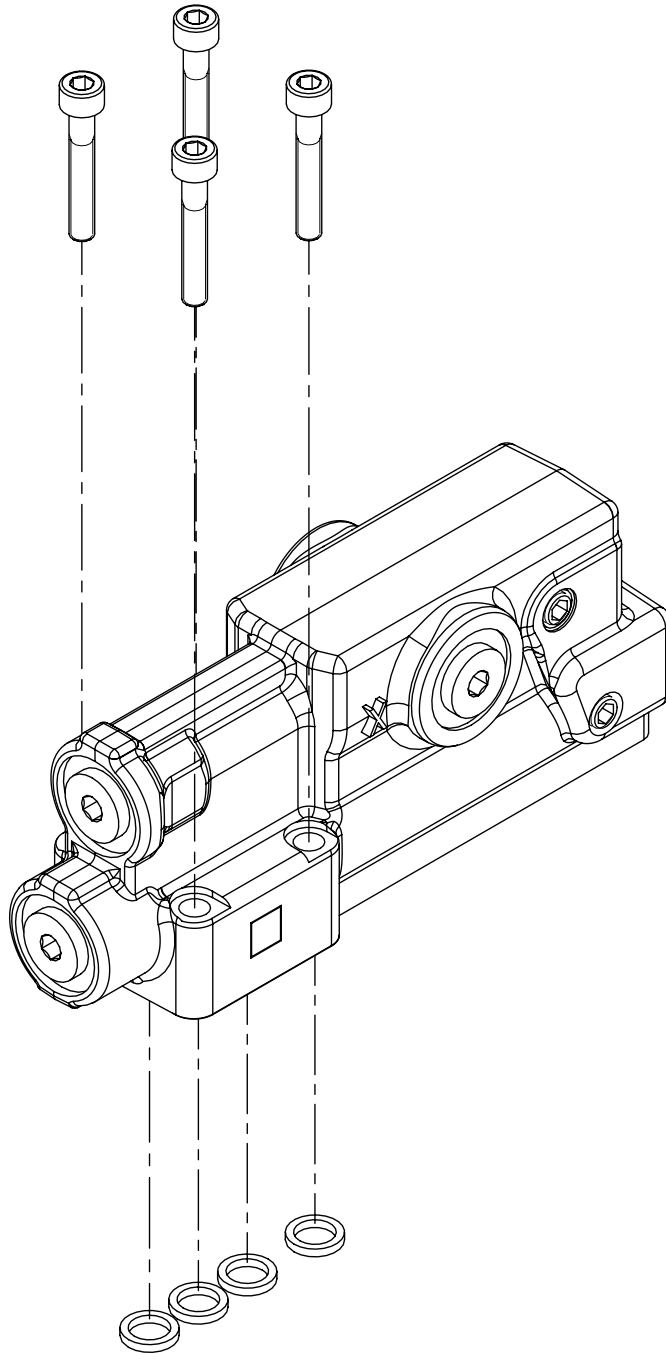
7. While holding the position of the PC adjustment screw, torque the PC set screw to 9 N·m [7 lbf·ft].
8. Stop the prime mover, remove the pressure gauges, and return the system to its normal operating configuration.

¹⁾ PC setting is referenced to case pressure. Subtract case pressure from system pressure to compute the actual setting.

4.3 Remote Pressure Compensator

A remote pressure compensation is realized by connecting a remote relief valve to the remote pressure port "X" through necessary plumbing. The required outlet pressure can be set by adjusting remote relief valve setting. Once pressure reaches preset value, the remote relief valve opens, resulting in Remote Pressure Compensator spool movement due to pressure imbalance. This will de-stroke the pump to maintain the set pressure.

Secondary pressure compensator is provided to limit the max. pressure setting as a fail-safe measure.



Ordering Code	Kit description	PC Setting range	RP Port type	Pump size
				28 cc
11421754	Controller kit	60bar - 100 bar	SAE	●
11421739	Controller kit	100 bar - 280 bar	SAE	●
11421740	Controller kit	280 bar - 310 bar	SAE	●
11421750	Controller kit	310 bar - 350 bar	SAE	●
11421741	Controller kit	60bar - 100 bar	ISO	●
11421767	Controller kit	100 bar - 280 bar	ISO	●
11421768	Controller kit	280 bar - 310 bar	ISO	●
11421751	Controller kit	310 bar - 350 bar	ISO	●
11421871	Controller seal kit	-	SAE	
11421924	Controller seal kit	-	ISO	●

4.3.1 Remote Pressure Compensator Setting

1. Install a pressure gauge to measure pump outlet pressure. Install a pressure gauge in drain port to measure case pressure. Tee-in a gauge to the RPC signal line.

WARNING

Escaping hydraulic fluid under pressure can have sufficient force to penetrate your skin causing serious injury and/or infection. Relieve pressure in the system before removing hoses, fittings, gauges, or components.

Unintended movement of the machine or mechanism may cause injury to the technician or bystanders. To protect against unintended movement, secure the machine or disable / disconnect the mechanism while servicing.

CAUTION

Contamination can damage internal components and void the manufacturer's warranty. Take precautions to ensure system cleanliness when removing and reinstalling system lines.

2. Start the prime mover and allow fluid to reach normal operating temperature. Slowly operate a hydraulic function which will demand approximately half flow from the pump but keep system pressure below the PC set point.
3. Loosen the RPC set screw. While watching the pressure gauges, turn the RPC adjustment screw until the desired pressure differential between the pump out pressure gauge and RPC port is achieved. Clockwise rotation will increase the setting; counterclockwise rotation will decrease; approximate gain = 12 bar [170 psi] per turn.
4. While holding the position of the RPC adjustment screw, torque the LS set screw to 9 N·m [7 lbf·ft].
5. Operate a hydraulic function to its full extension loading the pump at maximum pressure and zero flow.
6. Loosen the PC set screw and turn the PC adjustment screw until the desired setting is indicated on the pump outlet pressure gauge. Clockwise rotation will increase pressure; counterclockwise rotation will decrease; approximate gain = 35 bar [507 psi] per turn ¹⁾.

If the pressure does not increase, an external system relief valve may require adjustment. External system relief valve must be set above the PC setting for proper operation.

7. While holding the position of the PC adjustment screw, torque the PC set screw to 9 N·m [7 lbf·ft].
8. Stop the prime mover, remove the pressure gauges, and return the system to its normal operating configuration.

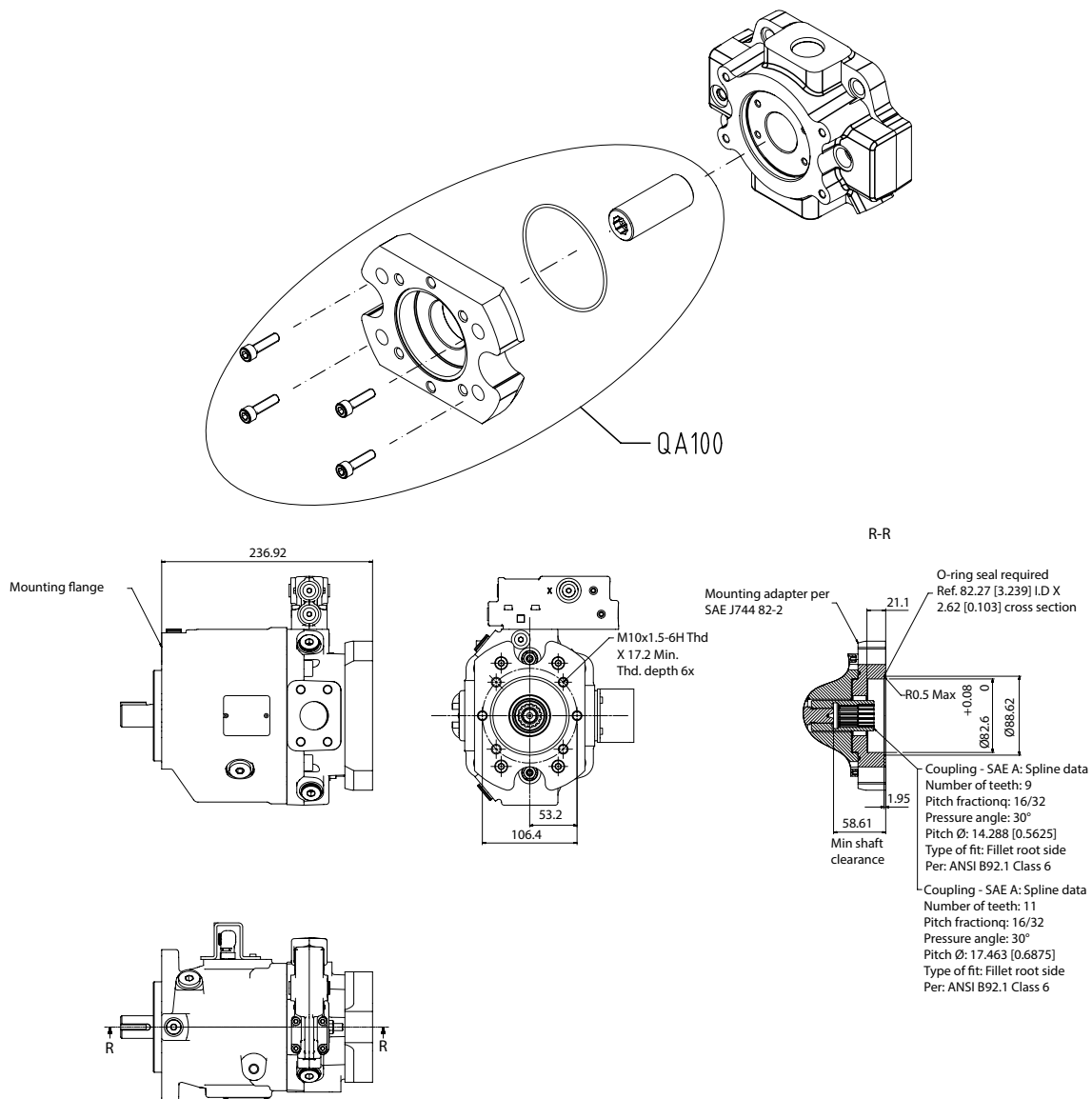
¹⁾ PC setting is referenced to case pressure. Subtract case pressure from system pressure to compute the actual setting.

5. Auxiliary mounting kits for tandem pump configuration

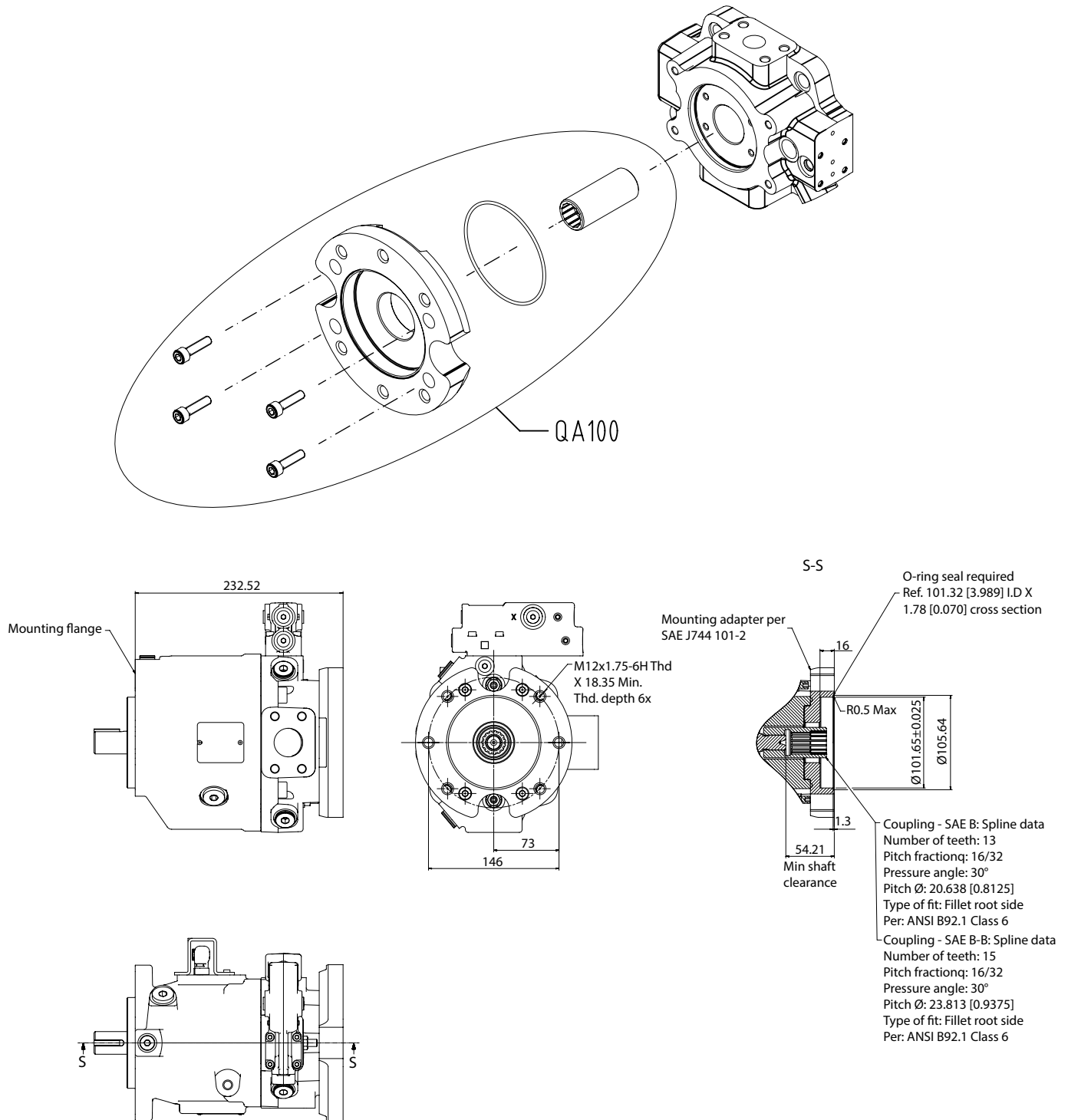
Auxiliary mounting kits meeting SAE and ISO specifications are available for all radially ported PVMX pumps enabling assembly of auxiliary hydraulic pumps. Oil from the main pump case lubricates the drive coupling.

Ordering Code	Auxiliary mounting specification	Pump size
		28 cc
11422143	SAE A, 2-Bolt, 9T Spline	●
11422182	SAE A, 2-Bolt, 11T Spline	●
11422177	SAE B, 2-Bolt, 13T Spline	●
11422191	SAE B, 2-Bolt, 15T Spline	●
11422198	ISO 3019/2-80A2HW 2-Bolt 9T Spline	●
11422237	ISO 3019/2-80A2HW 2-Bolt 11T Spline	●
11422265	ISO 3019/2-100A2HW 2-Bolt 13T Spline	●
11422282	ISO 3019/2-100A2HW 2-Bolt 15T Spline	●

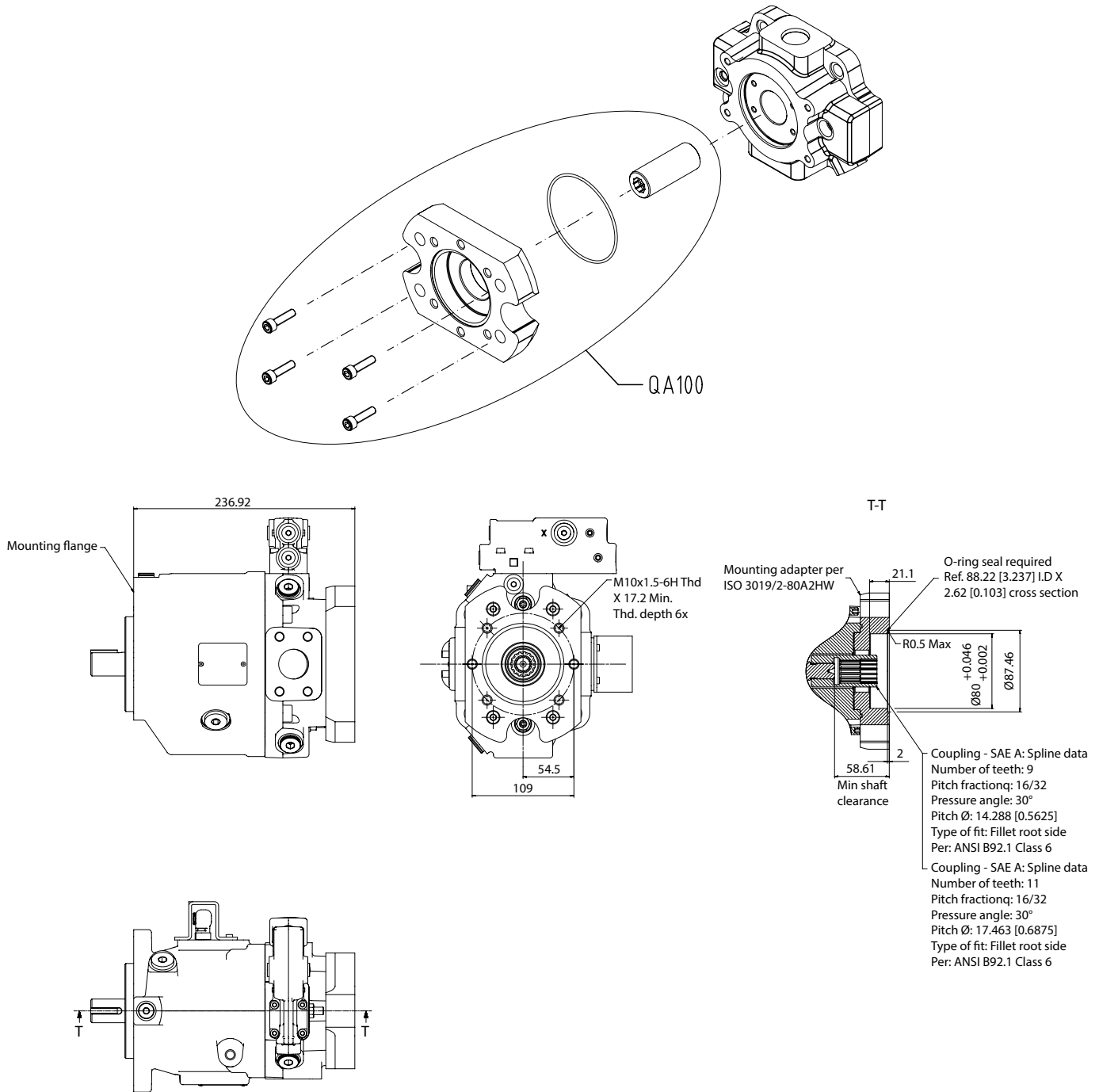
5.1 Auxiliary mounting kit SAE A, 2-Bolt, 9T and 11T



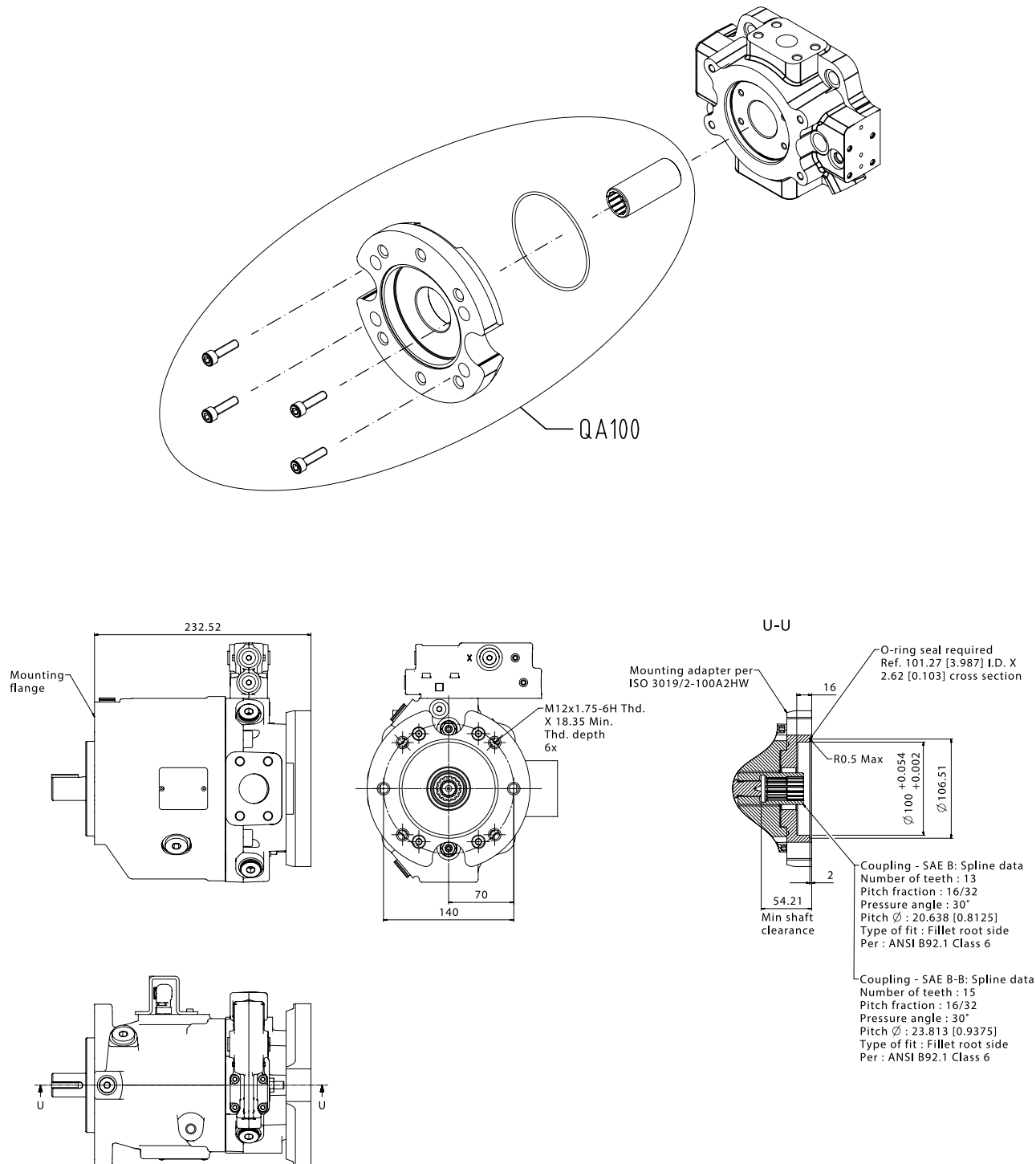
5.2 Auxiliary mounting kit SAE A, 2-Bolt, 13T and 15T



5.3 Auxiliary mounting kit ISO 3019/2-80A, 2-Bolt, 9T and 11T



5.4 Auxiliary mounting kit ISO 3019/2-80A, 2-Bolt, 13T and 15T



6. Initial start-up procedures

6.1 General

Follow this procedure when starting-up a new PVMX or when restarting an installation in which the pump has been removed.

WARNING

Unintended movement of the machine or mechanism may cause injury to the technician or bystanders. To protect against unintended movement, secure the machine or disable ! disconnect the mechanism while servicing.

Prior to installing the pump, inspect for damage incurred during shipping. Make certain all system components (reservoir, hoses, valves, fittings, heat exchanger, etc.) are clean prior to filling with fluid.

6.2 Start-up procedure

1. Connect the pump to the prime mover. Ensure that pump shaft is properly aligned with the shaft of the prime mover.

CAUTION

Incorrect shaft alignment may result in damage to drive shaft, bearings, or seal which can cause external oil leakage.

2. Fill the reservoir with recommended hydraulic fluid. Always filter fluid through a 10 micron filter pouring into the reservoir. Never reuse hydraulic fluid.
3. Fill the main pump housing with clean hydraulic fluid. Pour filtered oil directly into the upper most case drain port.
4. Fill the inlet line leading from the pump to the reservoir. Check the inlet line for properly tightened fittings and be certain it is free of restrictions and air leaks.
5. To ensure the pump stays filled with oil, install the case drain line in the upper most case drain port.
6. Install a gauge at port M2 (or equivalent system pressure measurement port if M2 not present) to monitor system pressure during start up.
7. While watching the pressure gauge installed at M2 or equivalent system pressure measurement port, jog the prime mover or run at the lowest possible speed until system pressure builds to normal levels (minimum 11 bar [160psi]). Once system pressure is established, increase to full operating speed. If system pressure is not maintained, shut down the prime mover, determine cause, and take corrective action.
8. Operate the hydraulic system for at least fifteen minutes under light load conditions.
9. Check and adjust control settings as necessary after installation.
10. Shut down the prime mover and remove the pressure gauge. Replace plug at port M2 or equivalent.
11. Check the fluid level in the reservoir; add clean filtered fluid if necessary.

The pump is now ready for operation.

7. Fluid and filter maintenance

To ensure optimum life, perform regular maintenance of the fluid and filter. Contaminated fluid is the main cause of unit failure. Take care to maintain fluid cleanliness when servicing.

Check the reservoir daily for proper fluid level, the presence of water, and rancid fluid odour. Water in the fluid may be noted by a cloudy or milky appearance or free water in the bottom of the reservoir. Rancid odour indicates the fluid has been exposed to excessive heat. Change the fluid immediately if these conditions occur. Correct the problem immediately.

Change the fluid more frequently if it becomes contaminated with foreign matter (dirt, water, grease, etc.) or if the fluid is subjected to temperature levels greater than the recommended maximum.

Dispose of used hydraulic fluid properly. Never reuse hydraulic fluid.

Change filters whenever the fluid is changed or when the filter indicator shows that it is necessary to change the filter. Replace all fluid lost during filter change.

8. Troubleshooting

Item	Description	Action
Check fluid level in reservoir.	Insufficient hydraulic fluid causes cavitation.	Fill the reservoir to proper level.
Check for air in system.	Air in system causes noisy, erratic control.	Purge air and tighten fittings. Check inlet for leaks.
Check pump inlet pressure/vacuum.	Improper inlet conditions cause erratic behavior and low output flow.	Correct pump inlet pressure/vacuum conditions. Refer to the Hydraulic Parameters topic.
Inspect shaft couplings.	A loose or incorrect shaft coupling causes excessive noise and/or vibration.	Repair or replace coupling and ensure that correct coupling is used.
Check shaft alignment.	Misaligned shafts create excessive noise and/or vibration.	Correct shaft misalignment.
Hydraulic fluid viscosity above acceptable limits.	Hydraulic fluid viscosity above acceptable limits or low fluid temperature will not allow the pump to fill or control to operate properly.	Allow system to warm up before operating, or use fluid with the appropriate viscosity grade for expected operating temperatures.

Actuator response is sluggish

Item	Description	Action
Check external system relief valve setting.	Low external relief valve setting slows down system.	Adjust external relief valve setting following manufacturer's recommendations. External relief setting must be above PC setting to operate properly.
Check PC and LS control setting.	Low PC setting prevents the pump from achieving full stroke. Low LS setting limits output flow.	Adjust PC and LS setting.
Check LS control signal pressures.	Incorrect LS signal will not allow pump to operate correctly.	Inspect system to ensure that proper LS signal transmit to pump.
Internal system leaks.	Worn internal parts don't allow the pump to operate properly.	Refer to Authorized Service Center for required repair.
Hydraulic fluid viscosity above acceptable limits.	Hydraulic fluid viscosity above acceptable limits or low fluid temperature will not allow the pump to fill or control to operate properly.	Hydraulic fluid viscosity above acceptable limits or low fluid temperature will not allow the pump to fill or control to operate properly.
Check external system valving.	Malfunctioning valving may not allow system to respond properly.	Repair or replace system valving as required.
Check pump case pressure.	High case pressure causes the system to be sluggish.	Correct case drain line restrictions.
Check pump inlet pressure/vacuum.	High inlet vacuum causes low output flow.	Correct inlet pressure conditions.

System operating hot

Item	Description	Action
Check fluid level in reservoir.	Insufficient volume of hydraulic fluid will not meet cooling demands of system.	Fill reservoir to proper level. Verify proper size of reservoir.
Hydraulic fluid viscosity above acceptable limits.	Fluid viscosity above acceptable limits or low fluid temperature will not allow the pump to fill or control to operate properly.	Allow system to warm up before operation or use fluid with the appropriate viscosity grade for expected operating temperatures.
Check external system relief valve setting.	Fluid passing through relief valve adds heat to system.	Adjust external system relief valve setting following manufacturer's recommendations. External relief valve setting must be above PC setting for proper operation.
Check PC and LS control setting.	Low PC setting will prevent the pump from achieving full stroke. Low LS setting will limit output flow.	Adjust PC and LS setting.
Check pump inlet pressure/vacuum.	High inlet vacuum adds heat to system.	Correct inlet pressure/vacuum conditions.
Check input speed.	Low input speeds decrease flow.	Adjust input speed.
Check pump rotation.	Incorrect rotational configuration will cause low flow.	Use pump with appropriate rotational configuration.

PVMX Open Circuit Variable Displacement Axial Piston Industrial Pumps

Low pump output flow

Item	Description	Action
Check fluid level in reservoir.	Insufficient hydraulic fluid will limit output flow and cause internal damage to pump.	Fill reservoir to proper level.
Hydraulic fluid viscosity above acceptable limits	Fluid viscosity above acceptable limits or low fluid temperature will not allow the pump to fill or control to operate properly.	Allow system to warm up before operating, or use fluid with the appropriate viscosity grade for expected operating temperatures.
Check external system relief valve setting.	External relief valve set below PC setting causes low output flow.	Adjust external relief valve following manufacturer's recommendation. External relief valve setting must be above PC setting to operate properly.
Check PC and LS control setting.	Low PC setting prevents the pump from achieving full stroke.	Adjust PC and LS setting.
Check pump inlet pressure/vacuum.	High inlet vacuum causes low output flow.	Correct inlet pressure conditions.
Check input speed.	Low input speeds decrease flow.	Adjust input speed.
Check pump rotation.	Incorrect rotational configuration will cause low flow.	Use pump with appropriate rotational configuration.

Pressure or flow instability

Item	Description	Action
Check for air in system.	Air in system causes erratic operation.	Activate PC allowing system to bleed air. Check inlet line for leaks and eliminate source of air ingress.
Check control spools.	Sticking control spools cause erratic operation.	Inspect spools for free movement in bore. Clean or replace.
Check LS setting.	Low LS setting may cause instability.	Adjust LS setting to proper level.
Check LS signal line.	Blocked LS signal line interferes with proper LS operation.	Remove blockage.
Check external relief valve and PC setting.	Insufficient pressure differential between PC setting and external relief valve.	Adjust external relief valve or PC control settings to appropriate level. Relief valve setting must be above PC setting to operate properly.
Check external relief valve.	Chattering external relief valve may cause unstable feedback to pump control.	Adjust or replace relief valve.

System pressure not reaching PC setting

Item	Description	Action
Check PC control setting.	System pressure will not rise above PC setting.	Adjust PC to appropriate setting.
Check external relief valve.	External relief valve setting below PC setting presents pressure compensation.	Adjust external relief valve according to manufacturer's recommendations. External relief valve must be set above PC setting to operate properly.
Inspect PC control spring.	Broken, damaged, or missing spring will cause erratic operation.	Replace the spring as required.
Inspect PC spool for wear.	Wear of PC spool causes internal leakage in the control.	Replace the spool as required.
Inspect PC spool for proper orientation.	Improper orientation results in poor operation.	Correct orientation of spool.
Check PC control for contamination.	Contamination may interfere with movement of the PC spool.	Clean PC control components, take appropriate action to eliminate contamination.



High inlet vacuum

CAUTION

High inlet vacuum causes cavitation which can damage internal pump components.

Item	Description	Action
Check fluid temperature.	Low temperature increases viscosity. High fluid viscosity causes high inlet vacuum.	Allow system to warm up before operating.
Inspect inlet screen.	Blocked or restricted inlet screen causes high inlet vacuum.	Clean screen/remove blockage.
Check inlet piping.	Too many fittings, bends, or long piping causes high inlet vacuum.	Eliminate fittings to make path more direct.
Hydraulic fluid viscosity above acceptable limits.	High fluid viscosity causes high inlet vacuum.	Select fluid with appropriate viscosity for expected operating temperature.



Cartridge valves

- DCV directional control valves
- Electric converters
- Electric machines
- Electric motors
- Fluid Conveyance
- Gear motors
- Gear pumps
- Hydraulic integrated circuits (HICs)
- Hydrostatic motors
- Hydrostatic pumps
- Industrial hydraulics
- Orbital motors
- PLUS+1® controllers
- PLUS+1® displays
- PLUS+1® joysticks and pedals
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